

Nonlocal corrections to surface plasmon-polaritons

This set of notes derives the equations reported in the paper Nonlocal response in thin-film waveguides: Loss versus nonlocality and breaking of complementarity

(<https://journals.aps.org/prb/pdf/10.1103/PhysRevB.88.115401>) by **Raza, Christensen, et al.**

Nonlocality is given through the relation between the current density and the electric field,

$$\frac{\beta_F^2}{\omega(\omega + i\eta)} \nabla(\nabla \cdot \mathbf{J}(\mathbf{r}, \omega)) + \mathbf{J}(\mathbf{r}, \omega) = \sigma(\omega) \mathbf{E}(\mathbf{r}, \omega) \rightarrow$$

$$\mathbf{J}(\mathbf{r}, \omega) = \sigma(\omega) \mathbf{E}(\mathbf{r}, \omega) - \frac{\beta_F^2}{\omega(\omega + i\eta)} \nabla(\nabla \cdot \mathbf{J}(\mathbf{r}, \omega)),$$

with $\sigma(\omega) = ine^2/m(\omega + i\eta) = i\varepsilon_0\omega_p^2/(\omega + i\eta)$ giving the (spatially local) free electron response.

Case of the single metal-insulator interface

We first recap Maxwell's equations,

$$\nabla \times \mathbf{E}(\mathbf{r}, \omega) = i\omega\mu_0\mathbf{H}(\mathbf{r}, \omega), \nabla \times \mathbf{H}(\mathbf{r}, \omega) = -i\omega\varepsilon_0\varepsilon_\infty\mathbf{E}(\mathbf{r}, \omega) + \mathbf{J}(\mathbf{r}, \omega),$$

where ε_∞ accounts for the bound ionic polarization response and the free electronic response is captured by the current density, $\mathbf{J}(\mathbf{r}, \omega)$. Note that by taking the divergence of the second equation, we obtain: $\nabla \cdot \mathbf{J}(\mathbf{r}, \omega) = i\omega\varepsilon_0\varepsilon_\infty \nabla \cdot \mathbf{E}(\mathbf{r}, \omega)$. Therefore, we may alternatively write:

$$\mathbf{J}(\mathbf{r}, \omega) = \sigma(\omega) \mathbf{E}(\mathbf{r}, \omega) - i\beta_F^2\varepsilon_0\varepsilon_\infty \nabla(\nabla \cdot \mathbf{E}(\mathbf{r}, \omega))/(\omega + i\eta)$$

Let's first consider a homogeneous free electron medium. If we consider longitudinal modes, i.e. $\mathbf{E}(\mathbf{r}, \omega) = e^{i\mathbf{q}\cdot\mathbf{r}}e_{\mathbf{q}}$, we obtain, for the current density,

$$\mathbf{J}(\mathbf{r}, \omega) = e^{i\mathbf{q} \cdot \mathbf{r}} \left[\sigma(\omega)(\mathbf{q}/|\mathbf{q}|) + i\beta_F^2 \varepsilon_0 \varepsilon_\infty (\mathbf{q}|\mathbf{q}|)(1/(\omega + i\eta)) \right]$$

Putting this back into Ampere's law and taking the divergence, we obtain,

$$\omega \varepsilon_0 \varepsilon_\infty |\mathbf{q}| + i|\mathbf{q}| \sigma(\omega) - \beta_F^2 \varepsilon_0 \varepsilon_\infty |\mathbf{q}|^3 / (\omega + i\eta) = 0$$

In the absence of spatial dispersion, this just gives us:

$\varepsilon_\infty + i\sigma(\omega)/\omega \varepsilon_0 = 0 \rightarrow \varepsilon_\infty - \omega_p^2/\omega(\omega + i\eta) = 0$, which is the familiar requirement that the dielectric function vanish. In the presence of dispersion, we obtain, instead:

$$\varepsilon_\infty - \omega_p^2/\omega(\omega + i\eta) - \beta_F^2 \varepsilon_\infty |\mathbf{q}|^2 / \omega(\omega + i\eta) = 0 \rightarrow |\mathbf{q}|^2 = (\omega/\beta_F^2)(\omega + i\eta)(\varepsilon_M/\varepsilon_\infty),$$

where $\varepsilon_M = \varepsilon_\infty - \omega_p^2/\omega^2$.

The $|\mathbf{q}|^2$ found here corresponds to the paper's definition of k_{NL}^2 , so we massage the relation above slightly in order to write it in the form of the paper:

$$k_{NL}^2 = \frac{\omega}{\beta_F^2} \left[\omega + i\eta \right] \left[1 - \frac{\omega_p^2}{\varepsilon_\infty \omega(\omega + i\eta)} \right] = \frac{1}{\beta_F^2} \left[\omega(\omega + i\eta) - \omega_p^2/\varepsilon_\infty \right],$$

which is consistent with the definition of k_{NL} on **Page 2** of the paper.

Note that we may also derive this quasistatically,

$$\begin{aligned} \mathbf{J}(\mathbf{r}, \omega) &= \sigma(\omega) \mathbf{E}(\mathbf{r}, \omega) - i\beta_F^2 \varepsilon_0 \varepsilon_\infty \nabla(\nabla \cdot \mathbf{E}(\mathbf{r}, \omega))/(\omega + i\eta) \rightarrow \\ i\omega \rho(\mathbf{r}, \omega) &= \sigma(\omega) \rho(\mathbf{r}, \omega)/\varepsilon_0 + i\beta_F^2 \varepsilon_\infty |\mathbf{q}|^2 \rho(\mathbf{r}, \omega)/(\omega + i\eta) \rightarrow \\ \omega(\omega + i\eta) &= \left[\omega_p^2 + \beta_F^2 \varepsilon_\infty |\mathbf{q}|^2 \right] \rightarrow |\mathbf{q}|^2 = \left[\omega(\omega + i\eta) - \omega_p^2 \right] / \beta_F^2 \varepsilon_\infty \end{aligned}$$

where we used $\nabla \cdot \mathbf{E}(\mathbf{r}, \omega) = \rho(\mathbf{r}, \omega)/\varepsilon_0$, $\nabla \cdot \mathbf{J}(\mathbf{r}, \omega) = i\omega \rho(\mathbf{r}, \omega)$

We now move to the case of an interface. Note that if we look at TM modes, with

$\mathbf{H}(\mathbf{r}, \omega) = e^{iqx - Q|z|} \mathbf{e}_y$, we have that

$$\nabla \times \mathbf{H}(\mathbf{r}, \omega) = (Q(z/|z|), 0, iq) e^{iqx - Q|z|} = -i\omega \varepsilon_0 \varepsilon_\infty \mathbf{E}(\mathbf{r}, \omega) + \mathbf{J}(\mathbf{r}, \omega)$$

We first consider the case where there is no spatial dispersion. In this case, we have $Q_1 = \sqrt{q^2 - \varepsilon_D(\omega/c)^2}$ in the insulator and $Q_2 = \sqrt{q^2 - \varepsilon_M(\omega/c)^2}$ in the metal. The requirement that the parallel component of the electric field remain unchanged across the boundary then gives us:

$\mathbf{E}_{||}(-i\omega\varepsilon_0\varepsilon_\infty + \sigma(\omega)) = -Q_2\mathbf{e}_x$ in the metal and $\mathbf{E}_{||}(\mathbf{r}, \omega)(-i\omega\varepsilon_0\varepsilon_D) = Q_1\mathbf{e}_x$ in the insulator. Therefore, we have: $\varepsilon_D/Q_1 = -(\varepsilon_\infty + i\sigma(\omega)/\omega\varepsilon_0)/Q_2 = -\varepsilon_M/Q_2$. Equivalently, $\varepsilon_D Q_2/\varepsilon_M Q_1 = -1$, which corresponds to **Equation 13** of the paper in the local limit.

Introducing nonlocality in the metal-insulator configuration

Now we examine corrections due to nonlocality. We still have:

$(Q(z/|z|), 0, iq)e^{iqx-Q|z|} = -i\omega\varepsilon_0\varepsilon_\infty\mathbf{E}(\mathbf{r}, \omega) + \mathbf{J}(\mathbf{r}, \omega)$. However, now the relationship between $\mathbf{J}$ and $\mathbf{E}$ is different. We also have an additional boundary condition, $\mathbf{J}(x, y, z = 0) \cdot \mathbf{e}_z = 0$

We consider a solution given by $\mathbf{E}(\mathbf{r}, \omega) = \frac{1}{\varepsilon_M}\nabla \times \mathbf{e}_y e^{iqx+Qz} + E_1\nabla e^{iqx+Q'z}$, where we must have $q^2 - Q'^2 = k_{NL}^2$. We automatically have continuity of the magnetic field's transverse component (The magnetic field will be given by $\mathbf{H} = -(i/\omega\mu_0) \times (\omega/c)^2 \mathbf{e}_y e^{iqx+Qz}$), We now need to ensure continuity of the parallel component of the electric field:

$$-Q_2/\varepsilon_M + iqE_1 = Q_1/\varepsilon_D$$

We impose the boundary condition on the current, which gives us

$$\begin{aligned}
\mathbf{e}_z \cdot \mathbf{J}(\mathbf{r}, \omega) = 0 &\rightarrow \sigma(\omega) \mathbf{E}(\mathbf{r}, \omega) \cdot \mathbf{e}_z - i\beta_F^2 \varepsilon_0 \varepsilon_\infty \partial_z (\nabla \cdot \mathbf{E}(\mathbf{r}, \omega)) / (\omega + i\eta) = 0 \\
&\rightarrow \sigma(\omega) \left[\frac{iq}{\varepsilon_M} + E_1 Q' \right] = -i\beta_F^2 \varepsilon_0 \varepsilon_\infty (k_{NL}^2 Q' E_1) / (\omega + i\eta) \rightarrow \\
i\omega_p^2 \varepsilon_0 \left[\frac{iq}{\varepsilon_M} + E_1 Q' \right] &= -i\beta_F^2 \varepsilon_0 \varepsilon_\infty \frac{1}{\beta_F^2} \left[\omega^2 - \omega_p^2 / \varepsilon_\infty \right] Q' E_1 \rightarrow \\
\omega_p^2 \left[\frac{iq}{\varepsilon_M} + E_1 Q' \right] &= -\varepsilon_\infty \left[\omega^2 - \omega_p^2 / \varepsilon_\infty \right] Q' E_1 \rightarrow iq\omega_p^2 / \varepsilon_M = -\varepsilon_\infty \omega^2 Q' E_1 \\
&\rightarrow iE_1 = q\omega_p^2 / Q' \varepsilon_\infty \varepsilon_M \omega^2 = q(\varepsilon_\infty - \varepsilon_M) / Q' \varepsilon_\infty \varepsilon_M
\end{aligned}$$

So we obtain: $iqE_1 = q^2(\varepsilon_\infty / \varepsilon_M - 1) \times (1 / \varepsilon_\infty Q')$

So the dispersion relation is given by:

$$\begin{aligned}
-1 &= (Q_1 \varepsilon_M / Q_2 \varepsilon_D) - iqE_1 (\varepsilon_M / Q_2) \rightarrow 1 = \\
&-(Q_1 \varepsilon_M / Q_2 \varepsilon_D) + q^2 (\varepsilon_\infty - \varepsilon_M) / (\varepsilon_\infty Q' Q_2)
\end{aligned}$$

This is exactly **Equation 12** of the paper.

Insulator-Metal-Insulator Nonlocal corrections

We first start by deriving the dispersion in the **absence of nonlocal corrections**. For $z > w/2$ (in the insulating material with a **local response**), we take the electric field to be given by the following:

$$\mathbf{E}(\mathbf{r}, \omega) = (1/\varepsilon_D) \nabla \times \mathbf{e}_y e^{iqx - Q_1 z} = (Q_1, 0, iq)(1/\varepsilon_D) e^{iqx - Q_1 z},$$

with $q^2 - Q_1^2 = \varepsilon_D (\omega/c)^2$. The corresponding magnetic field is given by

$$\begin{aligned}
(-i/\omega\mu_0) \nabla \times \mathbf{E}(\mathbf{r}, \omega) &= \mathbf{H}(\mathbf{r}, \omega) \rightarrow \text{(because the field is transverse)} \\
\mathbf{H}(\mathbf{r}, \omega) &= (i/\omega\mu_0 \varepsilon_D) \nabla^2 (\mathbf{e}_y e^{iqx - Q_1 z}) = -i\omega\varepsilon_0 \mathbf{e}_y e^{iqx - Q_1 z}
\end{aligned}$$

Within the metal ($|z| < w/2$), we take the electric field to be given by:

$$\mathbf{E}(\mathbf{r}, \omega) = (A/\varepsilon_M) \nabla \times \mathbf{e}_y e^{iqx} (e^{Q_2 z} \pm e^{-Q_2 z}) = (A/\varepsilon_M) e^{iqx} \left[\pm (Q_2, 0, iq) e^{-Q_2 z} + (-Q_2, 0, iq) e^{Q_2 z} \right],$$

where the $+$ sign corresponds to even modes and the $-$ sign corresponds to odd modes. Furthermore, $q^2 - Q^2 = \varepsilon_M (\omega/c)^2$. Boundary conditions on the tangential part of the magnetic field (evaluated at $z = w/2$) give us:

$$A(e^{Q_2 w/2} \pm e^{-Q_2 w/2}) = e^{-Q_1 w/2}$$

Boundary conditions on the tangential part of the electric field (also evaluated at $z = w/2$) give us:

$$(Q_1/\varepsilon_D) e^{-Q_1 w/2} = A(Q_2/\varepsilon_M) (\pm e^{-Q_2 w/2} - e^{Q_2 w/2}) \rightarrow$$

by using the fact that $e^{-Q_1 w/2} = A(e^{Q_2 w/2} \pm e^{-Q_2 w/2})$, we may write

$$(Q_1/\varepsilon_D) = -(Q_2/\varepsilon_M) [\tanh, \coth](Q_2 w/2) \rightarrow$$

$$[\tanh, \coth](Q_2 w/2) = -(\varepsilon_M Q_1/\varepsilon_D Q_2),$$

where the notation $[\tanh, \coth]$ means we take $\tanh$ for the upper sign (even mode) and $\coth$ for the lower sign (odd mode). Note that this result is consistent with **Equations 18a and 18b** of the paper in the local response limit.

Introduction of nonlocality in the insulator-metal-insulator configuration

Now we introduce nonlocal corrections, this amounts to adding in an additional, longitudinal, term to the electric field for $|z| < w/2$ (within the electron liquid):

$$\mathbf{E}_L(\mathbf{r}, \omega) = B \nabla (e^{iqx+Q'z} \mp e^{iqx-Q'z}) = B e^{iqx} \left[(iq, 0, -Q') e^{-Q'z} \mp (iq, 0, Q') e^{Q'z} \right],$$

where recall that Q' is determined from q via the longitudinal parameter, k_{NL} (and note that k_{NL} is just a function of frequency). The boundary condition on the magnetic field does not change. The boundary condition on the tangential component of the electric field is given by:

$$(Q_1/\varepsilon_D)e^{-Q_1w/2} = A(Q_2/\varepsilon_M)(\pm e^{-Q_2w/2} - e^{Q_2w/2}) + iBq(e^{-Q'w/2} \mp e^{Q'w/2})$$

We also require the following, which amounts to the normal component of the current vanishing at the boundaries:

$$\varepsilon_\infty \left[-BQ'(\pm e^{Q'w/2} + e^{-Q'w/2}) + (A/\varepsilon_M)(iq)(e^{Q_2w/2} \pm e^{-Q_2w/2}) \right] = iqe^{-Q_1w/2}$$

We can simplify this as:

$$\begin{aligned} \varepsilon_\infty \left[-2BQ'[\cosh(Q'w/2), -\sinh(Q'w/2)] + (iq/\varepsilon_M)e^{-Q_1w/2} \right] &= iqe^{-Q_1w/2} \rightarrow \\ iqe^{-Q_1w/2} \left[1/\varepsilon_\infty - 1/\varepsilon_M \right] &= -2BQ'[\cosh(Q'w/2), -\sinh(Q'w/2)] \\ \rightarrow B &= (iq/2Q'[\cosh(Q'w/2), -\sinh(Q'w/2)])e^{-Q_1w/2} \left[1/\varepsilon_M - 1/\varepsilon_\infty \right] \end{aligned}$$

Therefore, we have:

$$\begin{aligned} (Q_1/\varepsilon_D) &= -(Q_2/\varepsilon_M)[\tanh(Q_2w/2), \coth(Q_2w/2)] + \\ &\quad (q^2/Q')[\tanh(Q'w/2), \coth(Q'w/2)] \times \\ &\quad (1/\varepsilon_M - 1/\varepsilon_\infty) \end{aligned}$$

We rewrite this as:

$$-\varepsilon_M Q_1/\varepsilon_D Q_2 = \tanh(Q_2w/2) - (q^2/\varepsilon_\infty Q_2 Q') \tanh(Q'w/2)(\varepsilon_\infty - \varepsilon_M)$$

Equivalently,

$$\tanh(Q_2w/2) = -\varepsilon_M Q_1/\varepsilon_D Q_2 - (q^2/\varepsilon_\infty Q_2 Q') \tanh(Q'w/2)(\varepsilon_M - \varepsilon_\infty)$$

For odd modes, replace $\tanh \rightarrow \coth$. We note that this result is entirely consistent with **Equations 18a and 18b** of the paper.

Metal-Insulator-Metal Nonlocal corrections

For $z > w/2$, we take the electric field to be:

$$\mathbf{E} = (1/\varepsilon_M)\nabla \times \mathbf{e}_y e^{iqx-Q_2z} + B\nabla e^{iqx-Q'z} = (Q_2, 0, iq)(1/\varepsilon_M)e^{iqx-Q_2z} + B(iq, 0, -Q')e^{iqx-Q'z}$$

Within the insulator ($|z| < w/2$), we take the electric field to be given by:

$$\mathbf{E} = (A/\varepsilon_D)\nabla \times \mathbf{e}_y e^{iqx}(e^{Q_1z} \pm e^{-Q_1z}) = (A/\varepsilon_D)e^{iqx} \left[(Q_1, 0, iq)e^{-Q_1z} \pm (-Q_1, 0, iq)e^{Q_1z} \right],$$

where the $+$ is taken for even modes and the $-$ is taken for odd modes. Boundary conditions on the magnetic field give us:

$$A(e^{Q_1w/2} \pm e^{-Q_1w/2}) = e^{-Q_2w/2}$$

Boundary conditions on the parallel component of the electric field gives us:

$$(Q_2/\varepsilon_M)e^{-Q_2w/2} + B(iq)e^{-Q'w/2} = -(AQ_1/\varepsilon_D)(e^{Q_1w/2} \mp e^{-Q_1w/2}) = -(Q_1/\varepsilon_D)[\tanh, \coth](Q_1w/2)e^{-Q_2w/2},$$

where the notation $[\tanh, \coth]$ indicates that $\tanh$ should be taken for the upper sign and $\coth$ should be taken for the lower sign. And our last boundary condition is given by:

$$(iq\varepsilon_\infty/\varepsilon_M)e^{-Q_2w/2} - (BQ'\varepsilon_\infty)e^{-Q'w/2} = Aiq(e^{Q_1w/2} \pm e^{-Q_1w/2}) = iqe^{-Q_2w/2}$$

From the second equation, we have:

$$(Q_2/\varepsilon_M) + iqBe^{(Q_2-Q')w/2} = -(Q_1/\varepsilon_D)[\tanh, \coth](Q_1w/d)$$

And from the last equation we have:

$$(iq\varepsilon_\infty/\varepsilon_M) - Q'\varepsilon_\infty Be^{(Q_2-Q')w/2} = iq$$

We multiply the first equation by $Q'\epsilon_\infty$ and the second by iq and add the equations together:

$$-q^2\epsilon_\infty/\epsilon_M + Q_2Q'\epsilon_\infty/\epsilon_M = -q^2 - (Q_1Q'\epsilon_\infty/\epsilon_D)\tanh(Q_1w/2)$$

So we have:

$$-(\epsilon_D/\epsilon_\infty Q_1Q') \left[q^2(1 - \epsilon_\infty/\epsilon_M) + Q_2Q'\epsilon_\infty/\epsilon_M \right] = [\tanh, \coth](Q_1w/2)$$

We now rewrite the left hand side of this equation as:

$$-(\epsilon_D Q_2/\epsilon_M Q_1) \left[\left[q^2(\epsilon_M - \epsilon_\infty)/\epsilon_\infty Q_2Q' + 1 \right] \right]$$

We have thus proven the relations in the paper given by **Equations 15a and 15b**.

Equivalence of boundary conditions

In the above, we used the following additional boundary conditions interchangeably:

$$\epsilon_\infty(\mathbf{e}_z \cdot \mathbf{E}_M) = \epsilon_D(\mathbf{e}_z \cdot \mathbf{E}_D) \text{ and } \mathbf{J} \cdot \mathbf{e}_z = 0 \text{ at the boundary,}$$

where $\mathbf{E}_M$ and $\mathbf{E}_D$ are the electric fields in the metal and the dielectric insulator, respectively. Note that since $0 = \nabla \cdot (\nabla \times \mathbf{H}) = -i\omega\epsilon_0\epsilon_\infty \nabla \cdot \mathbf{E} + \nabla \cdot \mathbf{J}$, we have, immediately, by Gauss's law (if the normal component of the current vanishes just below the interface) that $\epsilon_\infty(\mathbf{e}_z \cdot \mathbf{E}_M) = \epsilon_D(\mathbf{e}_z \cdot \mathbf{E}_D)$. Therefore, the first boundary condition above is immediately satisfied if we impose $\mathbf{J} \cdot \mathbf{e}_z = 0$ at the boundary.

We should check that these boundary conditions are consistent with what we would get if we took the boundary condition from the current relation,

$$\frac{\beta_F^2}{\omega(\omega + i\eta)} \nabla(\nabla \cdot \mathbf{J}(\mathbf{r}, \omega)) + \mathbf{J}(\mathbf{r}, \omega) = \sigma(\omega)\mathbf{E}(\mathbf{r}, \omega),$$

We focus on the $\mathbf{e}_z$ component of this relation, and we also use the fact that this component of the current vanishes at the interface:

$$\frac{\beta_F^2}{\omega(\omega + i\eta)} \partial_z (i\omega \varepsilon_0 \varepsilon_\infty \nabla \cdot \mathbf{E}(\mathbf{r}, \omega)) = \frac{i\omega_p^2 \varepsilon_0}{\omega + i\eta} (\mathbf{e}_z \cdot \mathbf{E}(\mathbf{r}, \omega)) \rightarrow$$

$$\beta_F^2 \varepsilon_\infty \partial_z (\nabla \cdot \mathbf{E}) = \omega_p^2 (\mathbf{e}_z \cdot \mathbf{E})$$

We now use the fact that the longitudinal component of the electric field may be written as a gradient, $\nabla \Psi$ and the fact that $\nabla^2 \Psi = -k_{NL}^2 \Psi$, which gives us (using the definition of k_{NL}):

$$\omega^2 \varepsilon_M \partial_z \Psi = -\omega_p^2 (\mathbf{e}_z \cdot \mathbf{E}(\mathbf{r}, \omega))$$

And we use the fact that $-\omega_p^2/\omega^2 = \varepsilon_M - \varepsilon_\infty$ to write:

$$\varepsilon_M (\mathbf{e}_z \cdot \mathbf{E}^L(\mathbf{r}, \omega)) = (\varepsilon_M - \varepsilon_\infty) (\mathbf{e}_z \cdot \mathbf{E}(\mathbf{r}, \omega)),$$

where $\mathbf{E}^L$ is the longitudinal (curl-free) part of $\mathbf{E}$. Therefore, we have:

$$\varepsilon_\infty (\mathbf{e}_z \cdot \mathbf{E}^L) = (\varepsilon_M - \varepsilon_\infty) (\mathbf{e}_z \cdot \mathbf{E}^T) \leftrightarrow \varepsilon_\infty (\mathbf{e}_z \cdot \mathbf{E}) = \varepsilon_M (\mathbf{e}_z \cdot \mathbf{E}^T)$$

Note that this condition may equivalently be obtained through Maxwell's equations:

$$\left(\nabla \times (\nabla \times \mathbf{E}) \right)_z = (\omega/c)^2 \varepsilon_\infty E_z \rightarrow \varepsilon_M (\mathbf{e}_z \cdot \mathbf{E}^T) = \varepsilon_\infty (\mathbf{e}_z \cdot \mathbf{E})$$

At this point, we must show that $\varepsilon_M (\mathbf{e}_z \cdot \mathbf{E}^T) = \varepsilon_\infty (\mathbf{e}_z \cdot \mathbf{E})$ is equivalent to the previously imposed boundary condition, $\varepsilon_\infty (\mathbf{e}_z \cdot \mathbf{E}_M) = \varepsilon_D (\mathbf{e}_z \cdot \mathbf{E}_D)$. In order to do so, note that we have (only at the boundary):

$$(\nabla \times \mathbf{H}) \cdot \mathbf{e}_z = -i\omega \varepsilon_0 \varepsilon_M (\mathbf{e}_z \cdot \mathbf{E}^T)$$

Furthermore, since the parallel component of $\mathbf{H}$ is continuous (and the z component of its curl only contains derivatives parallel to the interface of the parallel component of $\mathbf{H}$), we have necessarily that $\mathbf{e}_z \cdot \mathbf{E}^T \varepsilon_M = \varepsilon_D \mathbf{E}^T \cdot \mathbf{e}_z$. QED.

